

College of Computer
The Department of Computer Science
CS 451 - 1445H (451)



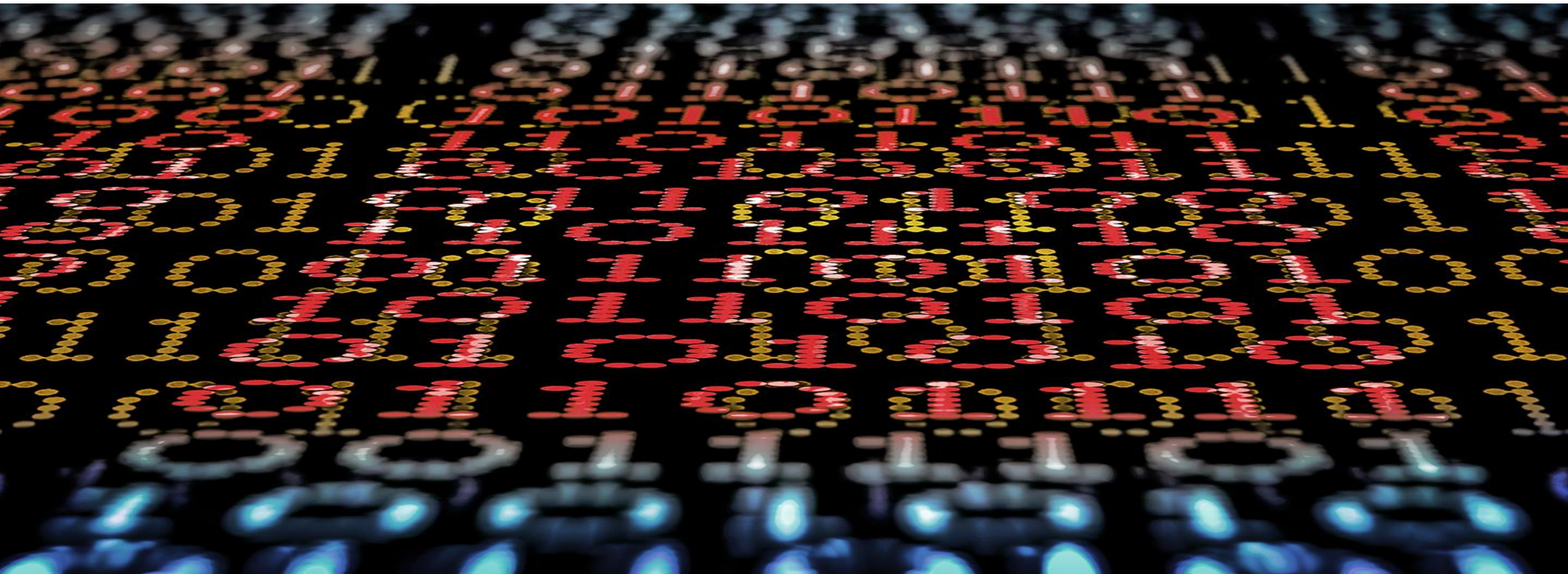
1091: Introduction to Computer Security (3 units)

Fall 2023

Wednesday 8:00 am to 10:35 am

15th Week

Lecture # 20



15

- **Data Security :**
- **Preparation: Think about how you could protect your Data ?**
- **Data Security :
(Cryptography)**
- **Preparation: How you can hide your information ?**
- **Reading: Provided Materials**

Data Security

- ◉ **Last Lecture :**

- ✓ **Firewalls**

- ✓ **Design of Firewalls**

- ✓ **Types of Firewalls**

- ✓ **Intrusion Detection and Prevention Systems**

- ✓ **Types of IDSs**

- ✓ **Intrusion Prevention Systems**

- ✓ **Intrusion Response**

Data Security

What is Data

- ◉ **Data is the raw material that is to be processed for information or for collection of details.**
- ◉ **It is unorganized data or facts that are to be processed**
- ◉ **Data is plain fact and it has to be processed for further information.**

Data Security

What is Data

- Because of this raw and possibly unorganized form, data may sometimes appear random, overly simple, or abstract.
- Think of data as the individual pieces of a jigsaw puzzle.



Data Security

What is Data

- ◉ **Data can be further broken down into both qualitative or quantitative.**
- ◉ **Qualitative data can be observed but not measured, and deals with aspects that may be observed by the senses**
- ◉ **Example: color, texture, smell, taste, appearance, etc.**

Data Security

What is Data

- Quantitative data is data that deals with numbers and can be measured.
- Criteria such as length, height, area, volume, weight, time, temperature, speed, cost, age, etc.

Data Security

What is Data

- Take, for example, a cup of coffee.



Data Security

Data in Computer world

- ◉ **Is the computers language.**
- ◉ **Data is the information that is input into the computer as quantities, characters, and symbols**
- ◉ **Then operations are performed on these data and stored as electrical signals then recorded on magnetic, optical, or mechanical recording media.**

Data Security

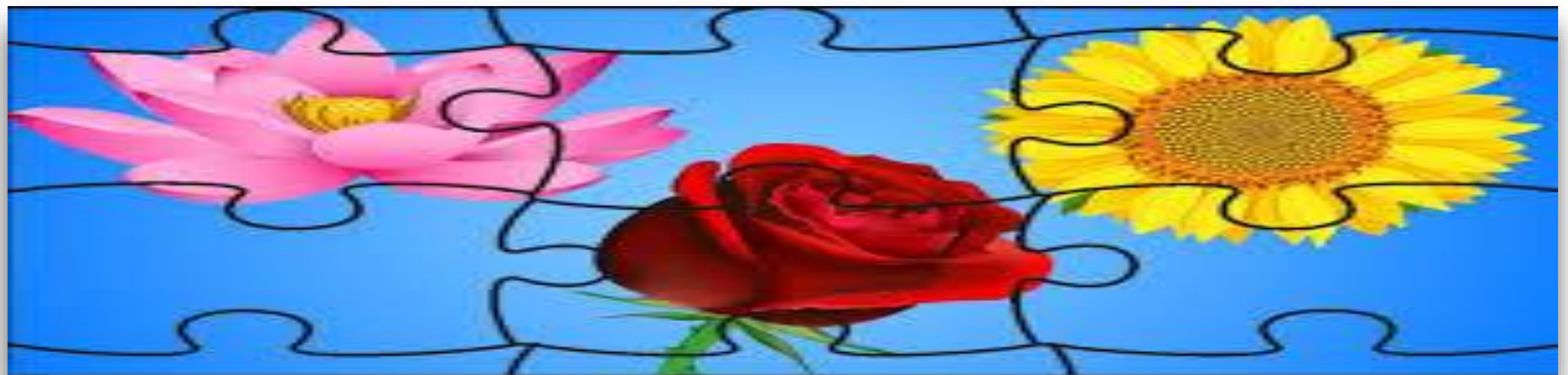
Data in Computer world

- ◉ **A computer program's component parts are just sets of data that are made up of coded software instructions that control the operations of a computer .**
- ◉ **It can be represented in many different ways, including tables, data trees, and graphs, among others.**

Data Security

What is Information

- Information is knowledge that has used and processed certain data and has rendered it useful.
- Information is the whole completed puzzle that the little data puzzle pieces helped you to put together.



Data Security

Data vs Information

- Without data there is no information – you can't put the puzzle together if there are no pieces, or if some of the pieces are missing.



Data Security

Data vs Information

- ◉ **Data is always correct – it is a tidbit of truth, a thing that has happened. However, information can be wrong.**
- ◉ **Information, like data, is a term that has applications when dealing with computers. If data are the tidbits that are put into the computer, it's information that comes out as a result.**

Data Security

Data vs Information

Data

Data is the input language for a computer

Data is unprocessed facts or mere figures

Data does not depend on information

Data is not specific

Data is the raw material that is collected

Information

Information is the output language for human.

Information is processed data which has been made sense of.

Information depends on data and without it, information cannot be processed.

Information is specific enough to generate meaning.

Information is a detailed meaning generated from the

Data Security

Data Classification

- ◉ **Public**
- ◉ **Sensitive**
- ◉ **Restricted**

Data Security

Data Status

- ◉ Data in motion
- ◉ Data at rest

Data Security

- ◉ **Data security is a set of standards and technologies that protect data from intentional or accidental destruction, modification or disclosure.**
- ◉ **Data security can be applied using a range of techniques and technologies**

Data Security

- ◉ **Physical security**
- ◉ **Administrative controls**
- ◉ **Logical controls**
- ◉ **Regulations and standard**
- ◉ **Other safeguarding techniques that limit access to unauthorized or malicious users or processes.**

Data Security

Different Data Security Technologies

- ◉ **Data encryption:** Data encryption applies a code to every individual piece of data and will not grant access to encrypted data without an authorized key being given
- ◉ **Data masking:** Masking specific areas of data can protect it from disclosure to external malicious sources, and also internal personnel who could potentially use the data. For example, the first 12 digits of a credit card number may be masked within a database.

Data Security

Different Data Security Technologies

- ◉ **Data erasure:** There are times when data that is no longer active or used needs to be erased from all systems.
- ◉ **Data resilience:** By creating backup copies of data, organizations can recover data should it be erased or corrupted accidentally or stolen during a data breach.

Cryptography

History of cryptography

- ***Cryptography* is a method of storing and transmitting data in a form that only those it is intended for can read and process.**
- **It is considered a science of protecting information by encoding it into an unreadable format.**
- **Cryptography is an effective way of protecting sensitive information as it is stored on media or transmitted through untrusted network communication paths.**

Cryptography

History of cryptography

- ◉ a goal of cryptography is to make obtaining the information too work-intensive or time-consuming to be worthwhile to the attacker.
- ◉ The first encryption methods date back to 4,000 years ago and were considered more of an art form.
- ◉ In Egypt, when hieroglyphics were used to decorate tombs to tell the life story of the deceased.

Cryptography

History of cryptography



Cryptography

History of cryptography

- ◉ **Atbash encryption scheme:** This method required the alphabet to be flipped so each letter in the original alphabet was mapped to a different letter in the flipped, or shifted, alphabet.
- ◉ **ABCDEFGHIJKLMNOPQRSTUVWXYZ**
- ◉ **ZYXWVUTSRQPONMLKJIHGFEDCBA**

Cryptography

History of cryptography

- ABCDEFGHIJKLMNOPQRSTUVWXYZ



- ZYXWVUTSRQPONMLKJIHGFEDCBA

- SECURITY → HVXFIRGB

- This is an example of *a substitution cipher*

Cryptography

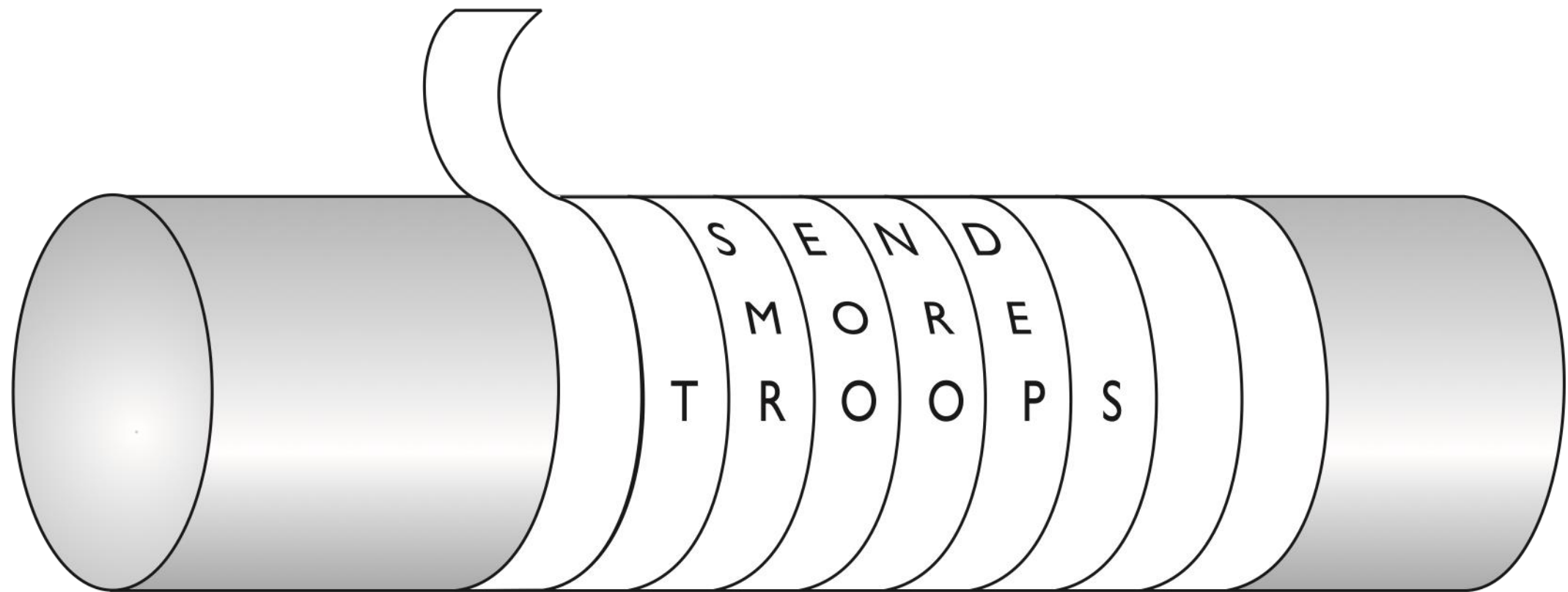
History of cryptography

- ◉ This type of substitution cipher is referred to as a *monoalphabetic substitution cipher* because it uses only **one alphabet**, whereas a *polyalphabetic substitution cipher* uses **multiple alphabets**.

Cryptography

History of cryptography

- **The Scytale cipher**



Cryptography

History of cryptography

- ◉ The **Julius Caesar method**:
- ◉ shifted the alphabet by three positions
- ◉ **ABC**DEFGHIJKLMNOPQRSTUVWXYZ
- ◉ DEFGHIJKLMNOPQRSTUVWXYZ**ABC**

Cryptography

History of cryptography

- ◉ During World War II
- ◉ The rotor cipher machine : Germany's Enigma machine



Cryptography

History of cryptography

- ◉ **When computers were invented, the possibilities for encryption methods and devices expanded exponentially and cryptography efforts increased dramatically.**
- ◉ **Lucifer : was developed at IBM**
- ◉ **In 1976: the U.S. Data Encryption Standard (DES)**

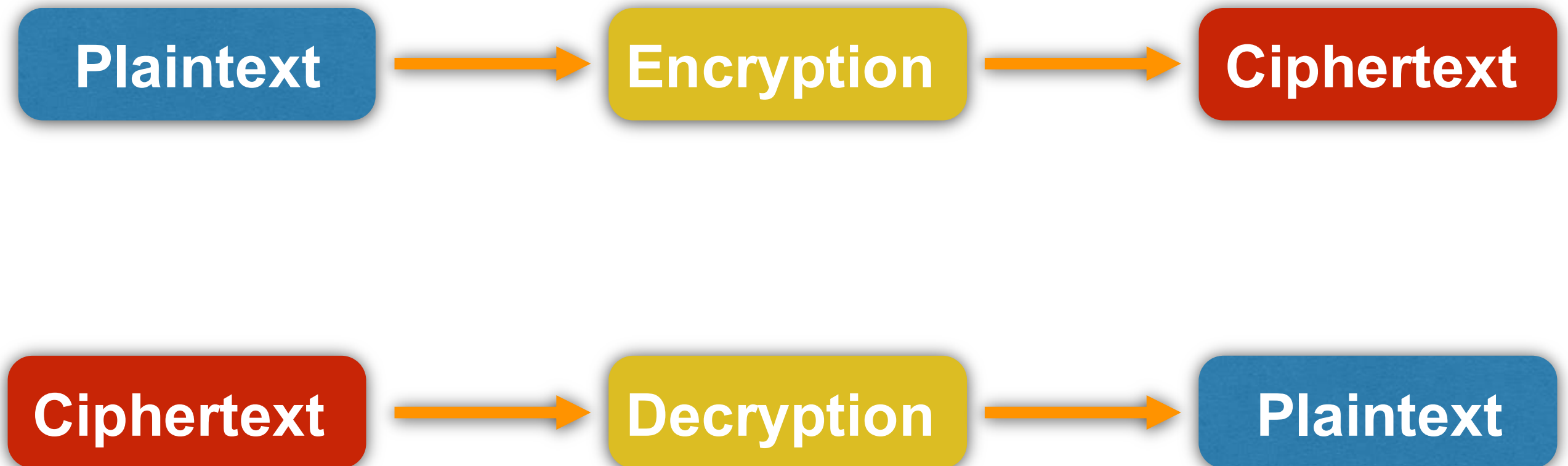
Cryptography

History of cryptography

- ◉ Encryption is used in hardware devices and in software to protect data
- ◉ **Cryptanalysis** is the science of studying and breaking the secrecy of encryption processes, compromising authentication schemes, and reverse-engineering algorithms and keys.

Cryptography

Cryptography Definitions and Concepts



Cryptography

Cryptography Definitions and Concepts

- **A cryptosystem** is a system or product that provides encryption and decryption
- This can be created through hardware components or program code in an application.
- The cryptosystem uses an **encryption algorithm** (which determines how simple or complex the encryption process will be), **keys**, and the necessary **software components and protocols**.

Cryptography

Cryptography Definitions and Concepts

- Breaking a cryptosystem can be accomplished by **a brute force attack**
- Important elements of encryption are to use **an algorithm without flaws**, **use a large key size**, **use all possible values within the keyspace**, and to protect the actual key. If one element is weak, it could be the link that dooms the whole process.

Cryptography

Cryptography Definitions and Concepts

- ◉ **Cryptosystems can provide the following services:**
 - **Confidentiality:** Renders the information unintelligible except by authorized entities.
 - **Integrity:** Data has not been altered in an unauthorized manner since it was created, transmitted, or stored.
 - **Authentication:** Verifies the identity of the user or system that created information.
 - **Authorization** Upon proving identity, the individual is then provided with the key or password that will allow access to some resource.
 - **Nonrepudiation** Ensures that the sender cannot deny sending the message.

Cryptography

Cryptography Definitions and Concepts

Cryptography Definitions

The following definitions are critical for your understanding of cryptography:

- **Access control** Restricting and controlling subject and object access attempts
- **Algorithm** Set of mathematical rules used in encryption and decryption
- **Cipher** Another name for algorithm
- **Cryptography** Science of secret writing that enables you to store and transmit data in a form that is available only to the intended individuals
- **Cryptosystem** Hardware or software implementation of cryptography that transforms a message to ciphertext and back to plaintext
- **Cryptanalysis** Practice of breaking cryptic systems
- **Cryptology** The study of both cryptography and cryptanalysis
- **Data origin authentication** Proving the source of a message (system-based authentication)
- **Encipher** Act of transforming data into an unreadable format
- **Entity authentication** Proving the identity of the entity that sent a message
- **Decipher** Act of transforming data into a readable format
- **Key** Secret sequence of bits and instructions that governs the act of encryption and decryption
- **Key clustering** Instance when two different keys generate the same ciphertext from the same plaintext
- **Keyspace** A range of possible values used to construct keys
- **Plaintext** Data in readable format, also referred to as cleartext
- **Receipt** Acknowledgment that a message has been received
- **Work factor** Estimated time, effort, and resources necessary to break a cryptosystem

If some of these terms do not make sense now, just hold on. We will cover them all in the following sections.

Cryptography

One-Time Pad

- ◉ ***A one-time pad* is a perfect encryption scheme because it is considered unbreakable if implemented properly.**
- ◉ **It was invented by Gilbert Vernam in 1917, so sometimes it is referred to as the Vernam cipher.**

Cryptography

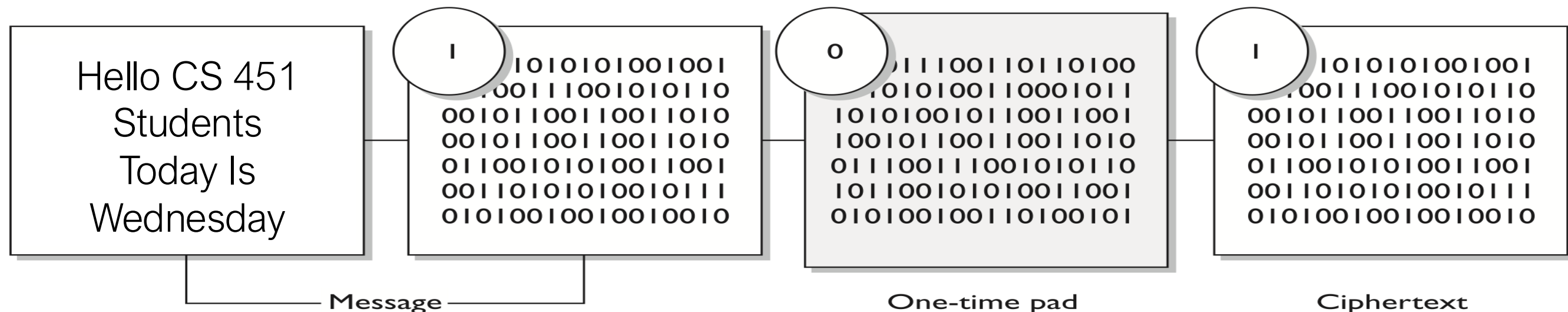
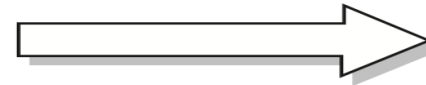
One-Time Pad

- ◉ **Message stream** 1001010111
- ◉ **Keystream** 0011101010
- ◉ **Ciphertext stream** 1010111101

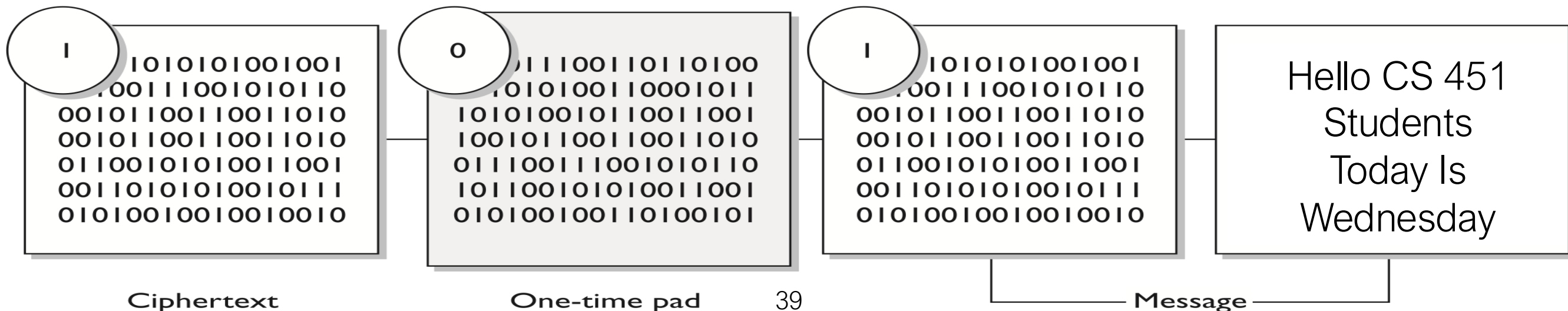
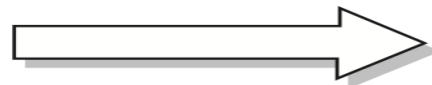
Cryptography

One-Time Pad

$$1 \text{ XOR } 0 = 1$$



$$1 \text{ XOR } 0 = 1$$



Cryptography

One-Time Pad

- ◉ **One-Time Pad Requirements**
- ◉ **For a one-time pad encryption scheme to be considered unbreakable, each pad in the scheme must be:**
 - **Made up of truly random values**
 - **Used only one time**
 - **Securely distributed to its destination**
 - **Secured at sender's and receiver's sites, at least as long as the message is there**

Cryptography

Running and Concealment Ciphers

- Two spy-novel-type ciphers are the running key cipher and the concealment cipher.
- The running key cipher :** uses components in the physical world around you
- A concealment cipher :** is a message within a message.

Cryptography

Steganography

- ◉ *Steganography* is a method of hiding data in another media type so the very existence of the data is concealed.
- ◉ Only the sender and receiver are supposed to be able to see the message because it is secretly hidden in image, video, document, or other type of media.
- ◉ Steganography is a type of security through obscurity.

Cryptography

Steganography components

- ◉ **Carrier** : A signal, data stream, or file that has hidden information (payload) inside of it
- ◉ **Stego-medium**: The medium in which the information is hidden
- ◉ **Payload** : The information that is to be concealed and transmitted

Cryptography

Types of Ciphers

- ◉ Symmetric encryption ciphers come in two basic types: **substitution** and **transposition (permutation)**.
- ◉ **The substitution** cipher replaces bits, characters, or blocks of characters with different bits, characters, or blocks.
- ◉ **The transposition** cipher does not replace the original text with different text, but rather moves the original values around , It rearranges the bits, characters, or blocks of characters to hide the original meaning.

Cryptography

Methods of Encryption

- ◉ The two main pieces are the **algorithms** and the **keys**
- ◉ **Algorithms** used in computer systems are complex mathematical formulas that dictate the rules of how the plaintext will be turned into ciphertext.
- ◉ A **key** is a string of random bits that will be used by the algorithm to add to the randomness of the encryption process.

Cryptography

Symmetric vs. Asymmetric Algorithms

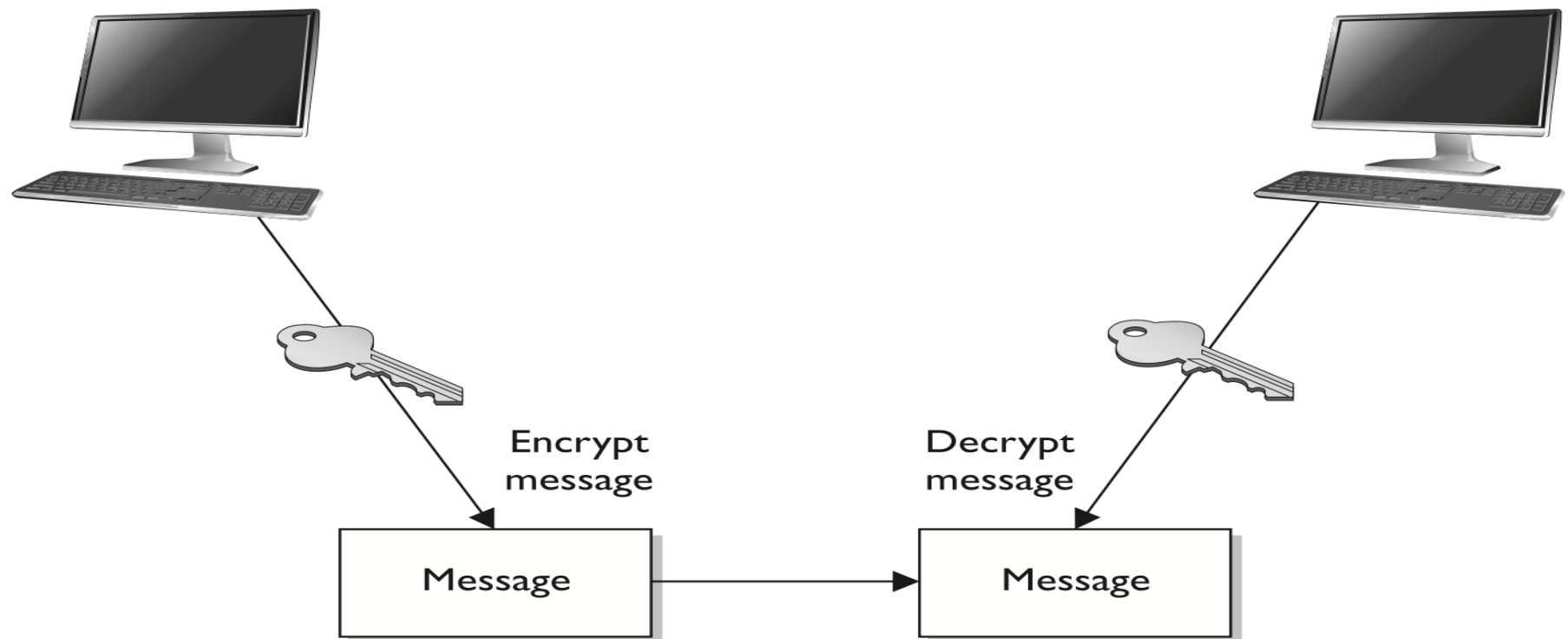
- Cryptography algorithms are either **symmetric algorithms, which use symmetric keys (also called secret keys)**, or **asymmetric algorithms, which use asymmetric keys (also called public and private keys)**

Cryptography

Symmetric Cryptography

- ◉ In a cryptosystem that uses symmetric cryptography, the sender and receiver use two instances of the same key for encryption and decryption,

Symmetric encryption uses the same keys.



Cryptography

Symmetric Cryptography

- ◉ So the key has dual functionality, in that it can carry out both encryption and decryption processes.
- ◉ Symmetric keys are also called *secret* keys, because this type of encryption relies on each user to keep the key a secret and properly protected.

Cryptography

Symmetric Cryptography

- ◉ **Because both users employ the same key to encrypt and decrypt messages, symmetric cryptosystems can provide confidentiality**
- ◉ **They cannot provide authentication or nonrepudiation.**
- ◉ **There is no way to prove through cryptography who actually sent a message if two people are using the same key.**

Cryptography

Strengths and weakness of symmetric key systems

- ◉ **Strengths:**

- **Much faster (less computationally intensive) than asymmetric systems.**
- **Hard to break if using a large key size.**

- ◉ **Weaknesses:**

- **Requires a secure mechanism to deliver keys properly.**
- **Each pair of users needs a unique key, so as the number of individuals increases, so does the number of keys, possibly making key management overwhelming.**
- **Provides confidentiality but not authenticity or nonrepudiation.**

Cryptography

Examples of symmetric algorithms

- ◉ The following are examples of symmetric algorithms
 - Data Encryption Standard (DES)
 - Triple-DES (3DES)
 - Blowfish
 - IDEA (International Data Encryption Algorithm)
 - RC4, RC5, and RC6
 - Advanced Encryption Standard (AES)

Cryptography

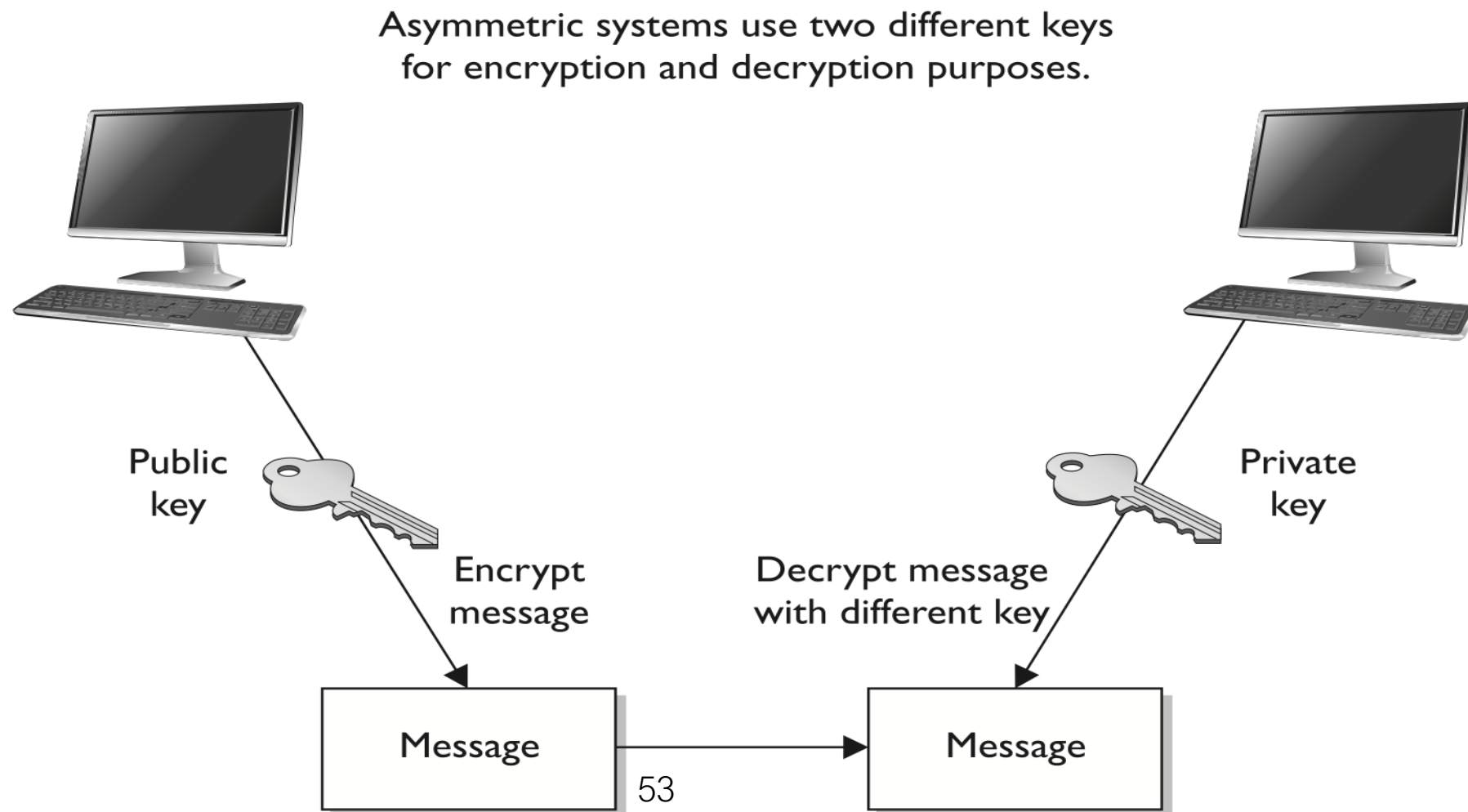
Asymmetric Cryptography

- ◉ In symmetric key cryptography, a single secret key is used between entities, whereas **in public key systems, each entity has different keys, or asymmetric keys.**
- ◉ The two different asymmetric keys are mathematically related.
- ◉ If a message is encrypted by one key, the other key is required in order to decrypt the message.
- ◉ The pair of keys is made up of one **public key** and one **private key**

Cryptography

Asymmetric Cryptography

- The *public key* can be known to everyone, and the *private key* must be known and used only by the owner.



Cryptography

Asymmetric Cryptography

- ◉ The *public key* can be known to everyone, and the *private key* must be known and used only by the owner.
- ◉ If we need confidentiality the sender would encrypt the data with the receiver's public key
- ◉ If we need authentication the sender would encrypt the data with the sender's private key

Cryptography

Strengths and Weaknesses of asymmetric key algorithms

◉ Strengths :

- Better key distribution than symmetric systems
- Better scalability than symmetric systems
- Can provide authentication and nonrepudiation

◉ Weaknesses :

- Works much more slowly than symmetric systems
- Mathematically intensive tasks

Cryptography

Examples of asymmetric key algorithms

- ◉ **RSA (Rivest-Shamir-Adleman)**
- ◉ **Elliptic curve cryptosystem (ECC)**
- ◉ **Diffie-Hellman**
- ◉ **El Gamal**
- ◉ **Digital Signature Algorithm (DSA)**
- ◉ **Merkle-Hellman Knapsack**

Cryptography

Differences between symmetric and asymmetric algorithms

Attribute	Symmetric	Asymmetric
Keys	One key is shared between two or more entities.	One entity has a public key, and the other entity has the corresponding private key.
Key exchange	Out-of-band through secure mechanisms.	A public key is made available to everyone, and a private key is kept secret by the owner.
Speed	Algorithm is less complex and faster.	The algorithm is more complex and slower.
Use	Bulk encryption, which means encrypting files and communication paths.	Key distribution and digital signatures.
Security service provided	Confidentiality.	Authentication and nonrepudiation.

Cryptography

Hybrid Encryption Methods

- ◉ **Asymmetric and Symmetric Algorithms Used Together**
- ◉ ***Public key cryptography* uses two keys (public and private) generated by an asymmetric algorithm for protecting encryption keys and key distribution**
- ◉ **A secret key is generated by a symmetric algorithm and used for bulk encryption.**
- ◉ **Then there is a hybrid use of the two different algorithms: asymmetric and symmetric.**
- ◉ **Each algorithm has its pros and cons, so using them together can be the best of both worlds.**

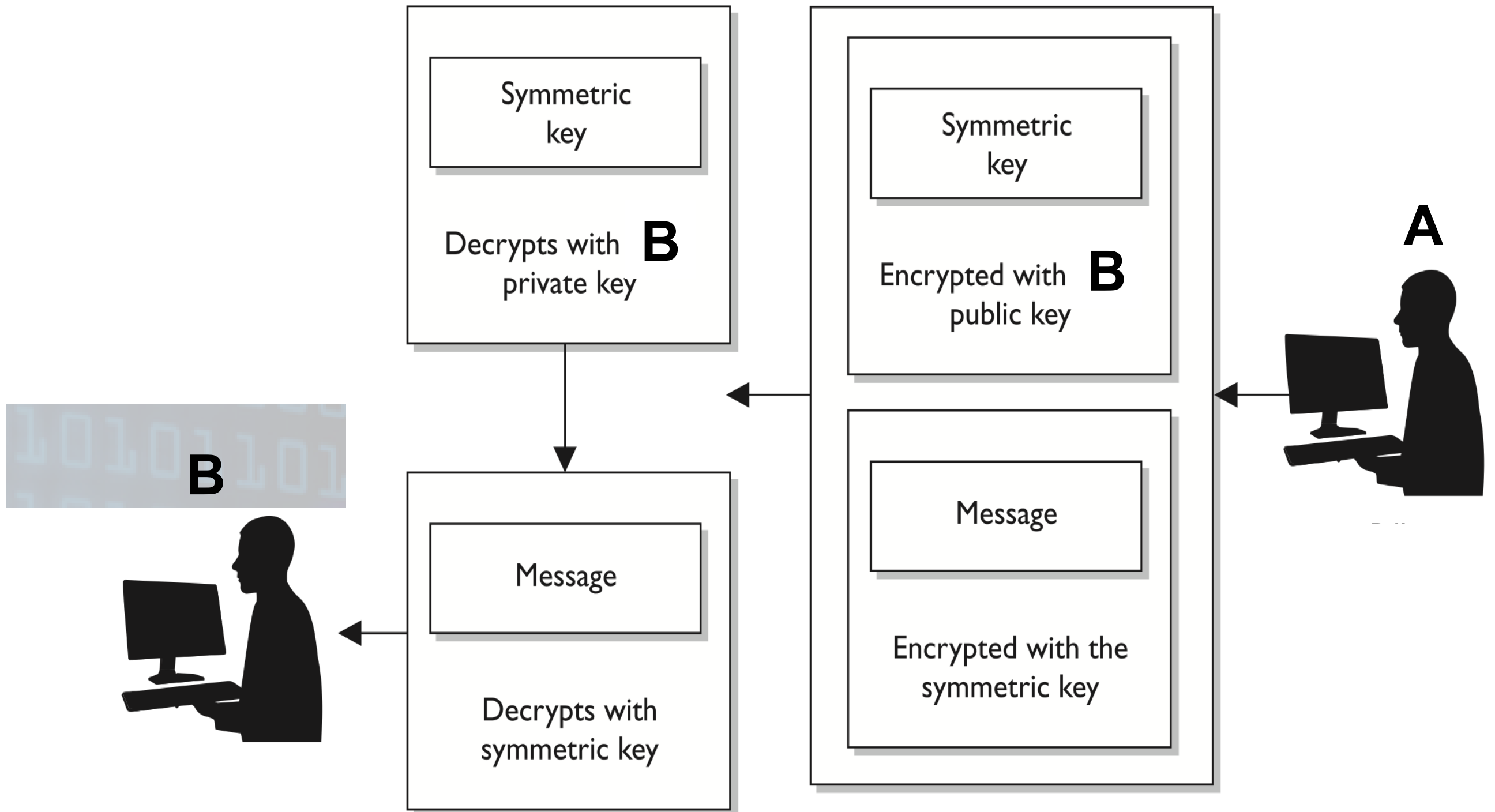
Cryptography

Hybrid Encryption Methods

- ◉ In the hybrid approach, the two technologies are used in a complementary manner, with each performing a different function.
- ◉ A symmetric algorithm creates keys used for encrypting bulk data, and an asymmetric algorithm creates keys used for automated key distribution.

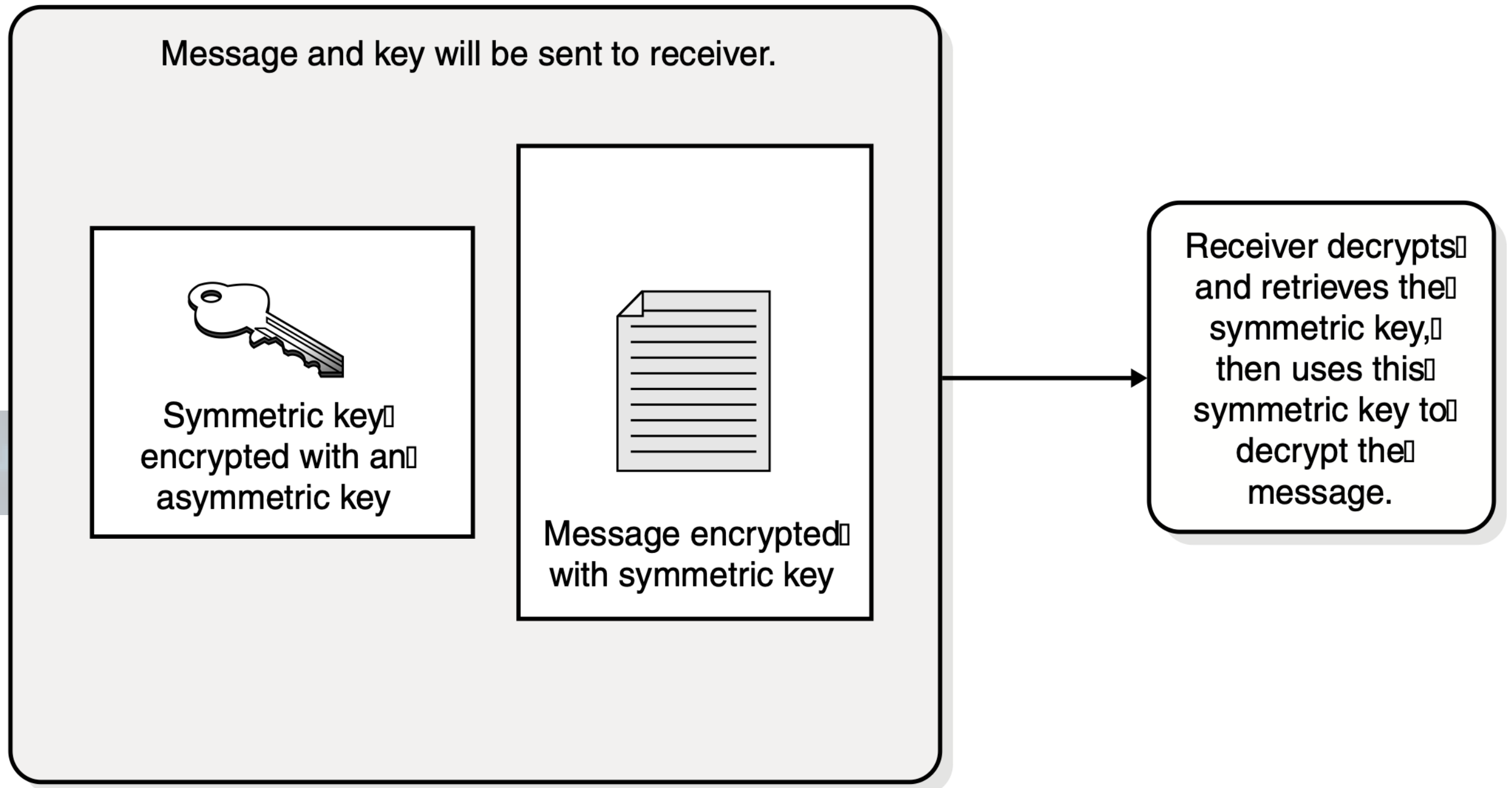
Cryptography

Hybrid Encryption Methods



Cryptography

Hybrid Encryption Methods



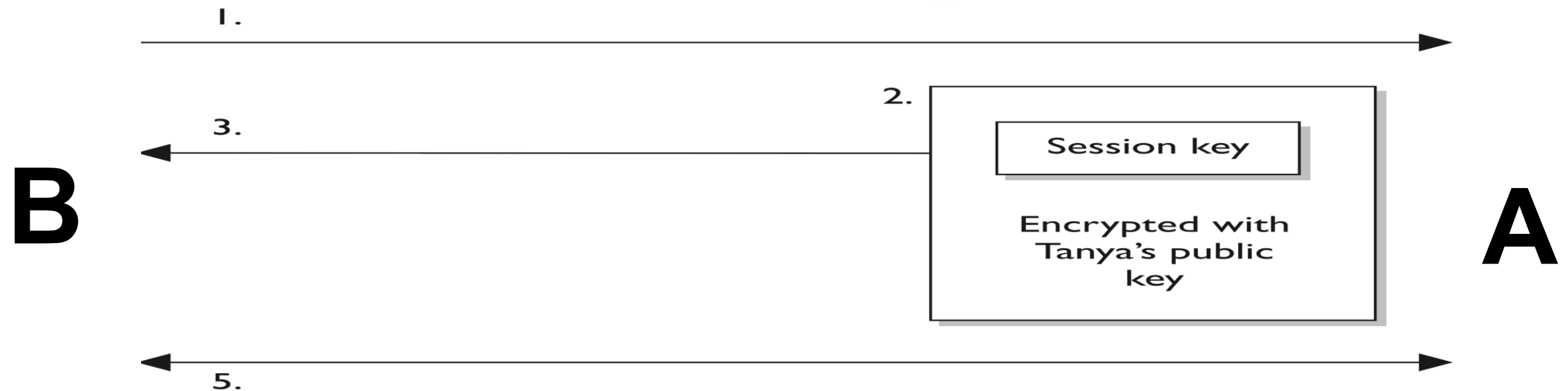
Cryptography

Session Keys

- ◉ ***A session key* is a single-use symmetric key that is used to encrypt messages between two users during a communication session.**

Cryptography

Session Keys



1. B sends A his public key

2. A generates a random session key and encrypts it using B's public key.

3. A sends the session key, encrypted with B's public key, to B.

4. B decrypts A's message with his private key and now has a copy of the session key.

5. B and A use this session key to encrypt and decrypt messages to each other.

Cryptography

Message Integrity

- ◉ **Hash algorithms:**

- **The One-Way Hash :** is a function that takes a variable-length string and a message and produces a fixed-length value called a hash value.
- **The hashing algorithm is not a secret—it is publicly known.**
- **The secrecy of the one- way hashing function is its “one-wayness.”**

The function is run in only one direction, not the other direction.

Cryptography

Application of Encryption

◉ **There are many applications for Cryptography in today's digital world :**

- **Digital Signatures**
- **Certificates**
- **E-mail**
- **Internet Security**

Cryptography

Attacks

- ◉ **Some active attacks that relate to cryptography :**
 - **Cipher-Only Attacks**
 - **Known-Plaintext Attacks**
 - **Chosen-Plaintext Attacks**
 - **Chosen-Ciphertext Attacks**
 - **Differential Cryptanalysis**
 - **Linear Cryptanalysis**

Cryptography

Attacks

- ◉ **Some active attacks that relate to cryptography :**
 - **Side-Channel Attacks**
 - **Replay Attacks**
 - **Algebraic Attacks**
 - **Analytic Attacks**
 - **Statistical Attacks**

3rd Assignment

- ◉ Why do we use **the symmetric key** on **the message** and **the asymmetric key** on **the symmetric key** ?
- ◉ If **a symmetric key** is encrypted with **a receiver's public key**, what security service(s) is provided?
- ◉ If data are encrypted with **the sender's private key**, what security service(s) is provided?

3rd Assignment

- ◉ Why do we encrypt the message with **the symmetric key**?
- ◉ Why don't we encrypt **the symmetric key** with another symmetric key?

Thank you

Q & A